

Sensor structure shapes the format of cognitive maps in navigation agents

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I. PROJECT RESCOPE AND CURRENT DIRECTION

The proposal framed the question around foveated vision and proposed a learned-gaze experiment (H3). Two deviations organise this report. **H1 reframed:** training the four sighted conditions under unified hyperparameters showed no foveated-specific compensatory effect, and a foveated-logpolar control landed in the rich-encoder regime as predicted by the encoder-bandwidth account; the substantive variable appears to be encoder bandwidth, not foveation. **H3 de-scoped:** the learned-gaze MLP collapsed under navigation reward, and we redirected effort to importing classical cognitive-neuroscience methods (Skaggs spatial information, SR-style predictive-horizon, Procrustes shape-metric, subspace divergence, excursion-forgetting) into the representational-analysis frame. The revised question: does visual-input structure shape the *format* of the cognitive map an agent builds in recurrent memory?

Capacity-allocation hypothesis A PointGoal navigation (PointNav) agent has two routes for spatial information: (i) recompute from the visual encoder each step, or (ii) integrate the GPS sensor across time in the LSTM memory. These routes compete for finite hidden-state capacity. A bottleneck leaves only the integration route, so the LSTM carries a temporally-stable, linearly-readable position code in the policy-readable layer (h_2). A rich encoder opens the visual route; capacity flows toward it and the linear component of h_2 shrinks while position stays recoverable non-linearly. Same task, different cognitive-map format [1], [2]. The hypothesis makes three falsifiable predictions tested below: linear h_2 GPS R^2 should fall with encoder bandwidth; a 2-layer MLP should still recover GPS; and lag- k stability should diverge between bottleneck and rich-encoder regimes.

II. SETUP AND METHOD

We hold architecture, training, and the non-visual sensor stack constant across five conditions and vary only the visual input (Table I). All agents share a 3-layer LSTM (512-d) and the goal-in-start-frame / GPS / compass / close-to-goal sensor stack of [3]. The four sighted conditions are trained from scratch under unified hyperparameters (250M frames); blind uses a 340M checkpoint, since blind agents need longer training to converge [4], [3]. All achieve 96–99% success and $SPL \in [0.85, 0.89]$ — no skill gap.

The conditions span the encoder spatial-bandwidth axis: *blind* is the no-vision lower bound [3]; *coarse* retains visual input but compresses it through a 1×1 encoder; *foveated-logpolar* is an intermediate-bandwidth (2×2) falsification target; *foveated* and *uniform* are rich-encoder endpoints (foveated has peripheral blur).

Table I
FIVE CONDITIONS, ORDERED BY ENCODER SPATIAL BANDWIDTH.

Condition	Spatial bandwidth	Visual input
Blind	none	no RGB
Coarse	1×1	48×48 RGB downsampled
Fov-logpolar	2×2	64×64 log-polar
Foveated	4×4	256^2 Gaussian-blur ($\sigma_{\max}=8$)
Uniform	4×4	256^2 full-resolution RGB

For each condition we collect 500 deterministic-policy episodes and analyse the frozen top-layer hidden state h_2 along three probing families [5]. **Decodability** (*what does h_2 encode?*) — linear (Ridge $\alpha=10$, what a linear policy head can read) and 2-layer MLP probes for GPS, lag- k probes for past position GPS_{t-k} , per-(unit, scene) Skaggs spatial information, and cross-training probes at 5 checkpoints. **Intrinsic structure** (*do conditions share a code?*) — linear CKA [6], with Procrustes shape-distance in progress. **Causal intervention** (*does the policy rely on h_2 ?*) — cross-condition memory transplant (sighted-only; coarse / blind pairs deferred). All probes use 5-fold episode-level cross-validation. This report focuses on decodability; the other two families are still in progress (§IV).

III. FINDINGS

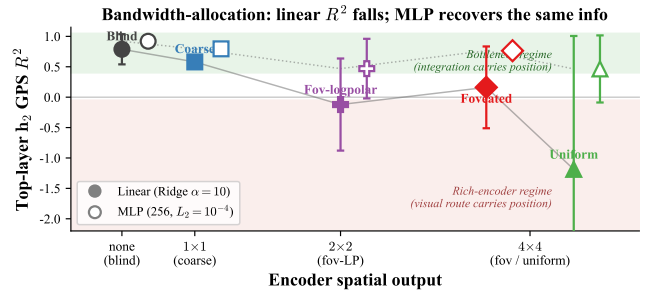


Figure 1. Linear (filled) and 2-layer MLP (open) probes for GPS, per condition. Linear R^2 falls monotonically with encoder spatial bandwidth, while the MLP probe recovers GPS in every condition: position information is retained in non-linear h_2 directions a linear policy head cannot read.

Encoder bandwidth shapes the linear axis Linear GPS R^2 falls smoothly as encoder spatial bandwidth grows (Figure 1, solid). The trend also serves as falsification: foveated-logpolar (a 2×2 rich-encoder control) was predicted to land in the rich-encoder regime, and the data are consistent with that prediction — arguing against the alternative that foveation specifically shapes the cognitive map. A second alternative — that the linear probe simply lacks capacity in rich-encoder states — is ruled out by a same-dimensional distance-to-goal control which decodes at $R^2 > 0.8$ in every condition. A 2-layer MLP probe on the same hidden

states recovers GPS everywhere (Figure 1, dashed), and the linear-MLP gap widens with encoder bandwidth: position information is retained, just not as the linearly-readable axis. The bottleneck leaves only the integration route, so position lands on a linearly-readable axis; a rich encoder opens a visual route and the integrated code stops being linearly readable, yet stays non-linearly recoverable. Format differs at convergence.

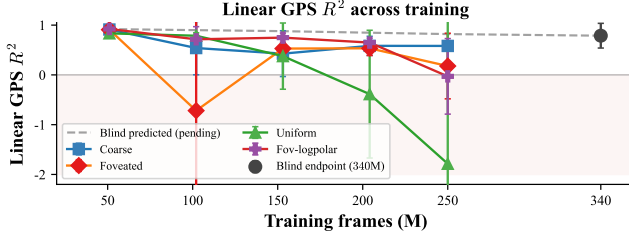


Figure 2. Linear GPS R^2 across training. All four sighted conditions emerge with a high linear top-layer GPS code at ~ 50 M frames; bottleneck (coarse) preserves the integrated code through training, while the three rich-encoder conditions decay as the visual route consolidates — decay rate ordered by encoder informativeness about position. Blind: 340M endpoint only (real point); the dashed line is the predicted trajectory under the capacity-allocation account, pending re-probing of intermediate blind checkpoints (§IV, A3).

Allocation emerges across training But did bottleneck conditions ever *learn* the visual route? Re-probing at intermediate checkpoints (Figure 2) suggests they did: all four sighted conditions begin with a high linear top-layer GPS code; the linear axis is the LSTM’s *default* early format. What separates conditions is what happens *after*: bottleneck preserves the integrated code; the three rich-encoder conditions decay as the visual route consolidates, with decay rate ordered by encoder informativeness about position. Capacity allocation thus appears to be a training-time process, not an endpoint property — bottleneck conditions kept the integration route on top because no usable visual alternative emerged.

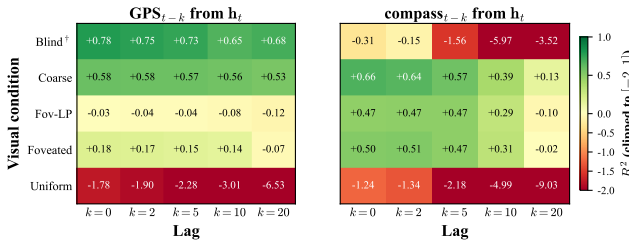


Figure 3. Past-target decoding from current $\mathbf{h}_t$ across lag k , for two spatial targets: GPS (left) and compass (right). Among the sighted conditions the bottleneck-vs-rich-encoder format contrast extends beyond GPS to compass — coarse sustains stable positive R^2 on both, while uniform degrades on both. Blind’s GPS profile fits the bottleneck pattern; its compass row is provisional under the in-progress probe-collect (§IV, A1) and not yet stable. (Negative- R^2 outliers clipped at -2.0 .)

Bottleneck integrates; rich encoder reacts The format difference also shows up within an episode and across two spatial targets. Decoding past GPS or compass from $\mathbf{h}_t$ (Figure 3) shows bottleneck conditions sustaining stable

positive R^2 across lags while rich-encoder conditions degrade: an *integrated* code preserves past spatial information; a *reactive* code keyed to current visual context does not. The contrast generalises across both targets — it appears to be about format, not GPS specifically. (Blind’s compass row is provisional pending the in-progress probe-collect.) As a check against the obvious null that bottleneck conditions just have more place cells, per-(unit, scene) Skaggs spatial information [3] is condition-flat at $[1.16, 1.35]$ bits — the bandwidth-allocation signal seems to live in linear readout rather than per-unit place-cell metric. Together the three findings suggest encoder bandwidth shapes the *format*, not the existence, of the cognitive map.

IV. OPEN ISSUES AND ACTION ITEMS

Forward risks Two scientific risks remain. The bandwidth-allocation account makes falsifiable predictions for several of the planned causal-intervention and scope-generalisation tests below; negative outcomes there would force a more nuanced reading of the encoder-memory race. Single-seed per condition also leaves seed-variance unbounded — the cross-condition rank ordering should survive but absolute magnitudes are seed-exposed.

Bandwidth-allocation refinements (A1) complete blind probe-collect and re-probe intermediate blind checkpoints, populating the blind row across all lenses (completes Figs 2–3); (A2) retrain coarse with 256×256 -padded input so the transplant matrix can include coarse pairs (closes the H2 evidence gap); (A3) two-seed validation on coarse and uniform endpoints to bound seed-variance.

New cog-neuro analyses on existing $\mathbf{h}_2$ (A4) allocentric occupancy decoder $f(\mathbf{h}_T, \mathbf{c}_T) \rightarrow 32 \times 32$ binary map — tests whether memory carries a metric *map* rather than just a position scalar (ground-truth ready); (A5) MINE-based predictive-horizon $I(\mathbf{h}_t; \text{pos}_{t+k})$ for $k \in \{1, 5, 10, 20\}$; (A6) successor-representation eigenbasis test linking $\mathbf{h}_2$ PCs to the eigenvectors of the policy’s successor matrix; (A7) Sussillo–Barak fixed-point analysis on $\mathbf{h}_2$; (A8) place-vs-view-cell dissociation, boundary-vector tuning, ratemap drift, and hierarchical scene-clustering as a relational map test.

Scope-generalisation (A9) foveation strength sweep ($\sigma_{\max} \in \{2, 4, 12, 20\}$) as a continuous lever on the encoder-memory race; (A10) transformer navigator under the same five conditions, probing attention / KV-cache for position — an architecture-agnostic falsification target; H3 reattempt with bounded- σ stochastic gaze if time permits.

V. IMPLICATIONS

Three implications follow. (i) Encoder bandwidth becomes a *design lever* for recurrent memory: restricting encoder output may recruit integration-style memory an unrestricted pipeline crowds out [7]. (ii) The encoder-memory race is architecture-agnostic: VLMs on rich foundation encoders may exhibit similar crowd-out, contributing to the spatial-reasoning gap [8]. (iii) The lens ports cognitive-neuroscience methodology to deep-RL.

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